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Exercise Performance With Aficamten vs Metoprolol in Obstructive Hypertrophic Cardiomyopathy
The MAPLE-HCM Randomized Clinical Trial

Gregory D. Lewis, MD; Pablo Garcia-Pavia, MD, PhD; Ahmad Masri, MD; Bela Merkely, MD; Michael E. Nassif, MD; Maria Luisa Peña-Peña, MD; Roberto Barriales-Villa, MD, PhD; Ozlem Bilen, MD; Melissa Burroughs, MD; Brian Claggett, PhD; Juan Pablo Costabel, MD; Edileide de Barros Correia, MD; Anne M. Dybro, MD, PhD; Ian J. Kulac, MS; Neal K. Lakdawala, MD; Amy Mann, BA; Martin S. Maron, MD; Shaina McGinnis, BS; Ajith Nair, MD; Chloe Newlands, MS; Steen H. Poulsen, MD; Patricia Reant, MD, PhD; P. Christian Schulze, MD; Scott D. Solomon, MD; Andrew Wang, MD; Regina Sohn, MD, PhD; Indrias Berhane, PhD; Stephen B. Heitner, MD; Daniel L. Jacoby, MD; Stuart Kupfer, MD; Fady I. Malik, MD, PhD; Amy Wohltman, ME; Michael A. Fifer, MD; Perry M. Elliott, MBBS, MD; for the MAPLE-HCM Investigators

IMPORTANCE Patients with obstructive hypertrophic cardiomyopathy (oHCM) endure life-altering exercise limitations. Current treatment guidelines recommend β-blockers as first-line therapy primarily based on expert opinion. The Metoprolol vs Aficamten in Patients with Left Ventricular Outflow Tract Obstruction on Exercise Capacity in HCM (MAPLE-HCM) trial characterizes comprehensive exercise response to aficamten monotherapy vs β-blockade (metoprolol).

OBJECTIVE To determine the effect of aficamten compared with metoprolol across all stages of exercise using 16 quantitative measures in individuals with oHCM.

DESIGN, SETTING, AND PARTICIPANTS This was a prespecified secondary analysis of the MAPLE-HCM study, a phase 3, randomized, active-control trial conducted from June 2023 to March 2025 with data analyzed between May and July 2025 at 71 sites in North America, South America, Europe, Israel, and China. Patients with symptomatic oHCM with objective evidence of exercise intolerance (peak oxygen uptake [pVO_2] <100% of predicted) were included. Those with a history of atrial fibrillation (paroxysmal or persistent), medical indication for β-blockers or calcium channel blockers prohibiting drug discontinuation, or intolerance or medical contraindication to β-blockers were excluded.

INTERVENTION Randomized 1:1 to titrated aficamten (5-20 mg daily) or matching titrated metoprolol (50-200 mg daily) for 24 weeks.

MAIN OUTCOMES AND MEASURES The main outcomes were submaximal exercise minute ventilation (VE)/carbon dioxide output (VCO_2) slope and anaerobic threshold; peak exercise duration, workload, heart rate and heart rate reserve; and postexercise oxygen recovery rates; and composite variables, such as circulatory power.

RESULTS Of 175 randomized participants (mean [SD] age, 57.7 [13.2] years; 102 [58.3%] male), 165 (94%) had core laboratory–validated exercise tests at baseline and week 24. Compared with metoprolol, aficamten treatment improved multiple stages of exercise; including submaximal exercise VE/VCO_2 slope (−2.8; 95% CI, −4.0 to −1.5; $P < .001$) and anaerobic threshold (76 mL/min; 95% CI, 41 to 111; $P < .001$); peak workload (8 watts; 95% CI, 3 to 13; $P = .003$); the time required for VO_2 to recover by 12.5% postexercise (−11 seconds; 95% CI, −16 to −5; $P < .001$); and circulatory power (819 mm Hg × mL/min per kg; 95% CI, 569 to 1070; $P < .001$). Large improvements (≥3.0 mL/kg/min) in peak VO_2 were more common with aficamten compared with metoprolol (17 [20.5%] vs 3 [3.7%], respectively; odds ratio, 6.8; 95% CI, 2.0 to 22.5; $P < .001$). In contrast, large reductions in peak VO_2 (≥3.0 mL/kg/min) were significantly more common with metoprolol than aficamten (17 [20.7%] vs 2 [2.4%], respectively; odds ratio, 10.6; 95% CI, 2.6 to ∞; $P < .001$).

CONCLUSIONS AND RELEVANCE This prespecified analysis of MAPLE-HCM demonstrated monotherapy with aficamten was superior to metoprolol in promoting adaptation to multiple phases of exercise in patients with symptomatic oHCM and supports aficamten as first-line therapy for oHCM.

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Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The MAPLE-HCM Investigators are listed in Supplement 4.

Corresponding Author: Gregory D. Lewis, MD, Mass General Brigham Heart and Vascular Institute, Bigelow 800, 55 Fruit Street, Boston, MA 02114 (glewis@mgb.org).

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The pathophysiology of hypertrophic cardiomyopathy (HCM) imposes multiple restraints on cardiac function during physical activity. The combination of dynamic left ventricular outflow tract (LVOT) obstruction, secondary mitral regurgitation and pulmonary hypertension, myocardial ischemia, chronotropic incompetence, reduced stroke volume reserve, and diastolic dysfunction collectively limit exercise tolerance in patients living with obstructive HCM (oHCM).[1-5]

β-Blockers have been first-line therapy for the management of symptomatic oHCM for more than 50 years following the demonstration that β-blockade attenuates LVOT gradient (LVOT-G),[5] and data from a 7-patient uncontrolled study[6] suggest that propranolol administration may increase exercise time and ameliorate exertional angina in oHCM. Aficamten is a next-in-class cardiac myosin inhibitor that inhibits formation of excessive actin-myosin cross bridges at the level of the sarcomere and mitigates cardiac hypercontractility. Aficamten treatment, when added to standard of care, including β-blockers, relieved obstruction and improved symptoms and exercise capacity.[7-9] However, a comparison of monotherapy with aficamten vs β-blockade on comprehensive exercise measures has not been reported.

In the Metoprolol vs Aficamten in Patients with Left Ventricular Outflow Tract Obstruction on Exercise Capacity in HCM (MAPLE-HCM) randomized clinical trial,[10] cardiopulmonary exercise testing (CPET) was used to measure breath-by-breath gas exchange parameters throughout all stages of exercise from onset to postexercise recovery. CPET captures oxygen uptake (VO_2) and ventilatory variables that have consistently been shown to predict prognosis in HCM and other cardiac disease,[11] and permits derivation of composite variables and novel measures of recovery that reflect cardiac performance.[12]

In MAPLE-HCM, we hypothesized that aficamten would improve 16 prespecified CPET measures of exercise performance that have not been previously compared between cardiac myosin inhibitor and β-blocker monotherapy for oHCM. We further assessed responder analysis for clinically meaningful improvements in peak VO_2 (pVO_2) and changes in oHCM disease severity in correlation with changes in pVO_2.

Key Points

Question How do the effects of aficamten compare to the effects of metoprolol across comprehensive exercise measures in obstructive hypertrophic cardiomyopathy (oHCM)?

Findings In the phase 3 MAPLE-HCM randomized clinical trial, 175 patients with symptomatic oHCM received aficamten or metoprolol monotherapy for 24 weeks. Aficamten and metoprolol had divergent effects on exercise performance that favored aficamten across all phases of exercise compared to metoprolol.

Meaning Aficamten treatment favorably impacted a broad array of exercise measures in comparison to metoprolol in this study and merits consideration for first line therapy for patients with symptomatic oHCM.

Methods

Trial Design and Oversight

The study (eFigure 1 in Supplement 3) was designed and executed by an academic steering committee in conjunction with the study sponsor. All study participants provided written informed consent prior to enrollment, and the study was carried out in accordance with the provisions of the Declaration of Helsinki and the International Council for Harmonization Guideline for Good Clinical Practice. An independent data monitoring committee had access to unblinded data for monitoring. Study personnel and participants remained blinded to treatment assignments, dosing, N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels, and echocardiogram measures through database lock. Analyses were conducted according to a prespecified statistical analysis plan (Supplement 2) that was finalized prior to locking the database. The trial was approved by the regulatory agencies in the participating countries and by the institutional review board or ethics committee at each trial center. The study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline.

The rationale for and design of MAPLE-HCM have been previously described.[13] In brief, this was a phase 3, randomized, double-dummy, double-blind multicenter international trial in participants with symptomatic oHCM. Individuals with LVOT-G ≥30 mm Hg (resting) or ≥50 mm Hg (during the Valsalva maneuver), New York Heart Association (NYHA) functional class II-III symptoms, baseline pVO_2 <100% of predicted,[14] and respiratory exchange ratio ≥1.05 were eligible to participate. Those with a history of atrial fibrillation (paroxysmal or persistent), medical indication for β-blockers or calcium channel blockers prohibiting drug discontinuation, or intolerance or medical contraindication to β-blockers were excluded. The database was locked on May 1, 2025, and unblinded thereafter.

Randomization

Participants who met screening criteria were randomly assigned 1:1 to receive either double-dummy treatment with aficamten plus placebo or metoprolol plus placebo. Randomization was performed using an interactive web-response system and stratified by CPET modality (treadmill or upright cycle ergometry), duration of oHCM diagnosis, and previous pharmacologic therapy exposure.

Interventions

Beginning at screening visit 1, eligible participants receiving standard-of-care therapies for HCM underwent washout of standard-of-care medications, which were discontinued 7 or more days prior to screening assessments. The starting doses of daily aficamten and metoprolol were 5 mg and 50 mg, respectively with subsequent dose escalation to a maximum dose of 20 mg and 200 mg, respectively. Titration was done in accordance with a prespecified algorithm based on data collection by personnel not otherwise involved in study conduct.[13]

Cardiopulmonary Exercise Testing

The CPET core laboratory (Massachusetts General Hospital) certified study sites in accordance with the CPET manual of

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operating procedures (version 4.0; Supplement 1). All CPET raw data were transferred electronically to the core laboratory, where they were interpreted in a blinded fashion. Prespecified criteria for valid CPETs included the absence of equipment malfunction, major CPET protocol deviation, or transient illness or injury unrelated to HCM symptoms that precluded CPET completion.

CPET End Points

The following prespecified exercise end points, with the exception of pVO_2, were derived and examined specifically for this substudy (eTable 1 in Supplement 3): (1) submaximal exercise response patterns (ie, ventilatory efficiency [VE/VCO_2 slope], VO_2 at the anaerobic threshold), derived from the V-slope method, and the extent to which work was performed aerobically during exercise, as captured by the VO_2/W slope throughout exercise; (2) measures of peak exercise response, including pVO_2, and assessments independent of gas exchange measures, such as peak workload, total exercise duration, and heart rate (HR) reserve; (3) composites that integrate more than 1 gas exchange variable or a combination of gas exchange and hemodynamic measures, including integrated submaximal and peak exercise response (equally weighted *z* score of VE/VCO_2 slope and pVO_2, circulatory power, and ventilatory power). This *z* score reflects CPET variables known to be closely associated with prognosis in HCM that also reflect distinct phases of exercise and was derived by reversing the directionality of VE/VCO_2 slope such that increases in both *z* score components indicate benefit, and equal weights were used for each component. Circulatory power (pVO_2 × peak systolic blood pressure), and ventilatory power (ratio of systolic blood pressure at peak exercise to VE/VCO_2 slope derived from all of exercise) were also calculated.[15]

Finally, we measured VO_2 kinetics during postexercise recovery, which provide insight into metabolic resilience following exercise and have been shown to track closely with degree of impairment in exercise hemodynamic responses.[13] We assessed the time to any decrement in VO_2 more than 0%, compared with peak values, which has been previously characterized as VO_2 recovery delay,[16] as well as the time to VO_2 decrements of 12.5%, 25%, and 50% after exercise. Spearman correlations adjusted for baseline values were derived for changes in pVO_2 relative to changes in other important clinical measures.

Sample Size Calculation

The sample size calculation for MAPLE-HCM assumed a between-group difference in change from baseline in pVO_2 of 2 mL/kg/min with treatment with aficamten vs metoprolol, a common SD in pVO_2 of 3 mL/kg/min, and 10% missing data for the primary end point. A sample size of 170 participants randomized 1:1 to aficamten or metoprolol was estimated to provide ≥90% power to detect a 2-mL/kg/min difference in pVO_2 change from baseline to week 24 with a 2-sided type I error of 0.05. Variance in repeated CPET measures other than pVO_2 have been shown to mirror variance in pVO_2, inferring similar power to detect differences of comparable magnitude in other CPET measures.[17]

Statistical Analysis

The full statistical analysis plan for CPET analyses is provided in Supplement 2. CPET efficacy end points were evaluated using an analysis of covariance model that included terms of treatment, randomization stratification factors (CPET modality, duration of oHCM diagnosis and pretrial pharmacologic therapy), and baseline value of the outcome as covariates. The full analysis set was used for efficacy analyses, which included all randomized participants who received 1 or more dose of the study drug and had 1 or more postbaseline efficacy assessment, unless otherwise specified. Least-squares mean treatment difference and standard errors were used to produce estimates of the treatment differences, their 95% CIs, and *P* values for the test of null hypothesis of no treatment effect.

Additional analyses included a responder analysis for the proportional achievement of clinically meaningful thresholds for shifts in pVO_2 (small [0 to <1.5 mL/kg/min], moderate [≥1.5 to <3], or large [≥3.0]) by treatment group and correlations between change in pVO_2 and other measures of treatment effect (symptoms, hemodynamics, biomarkers); and evaluation of how change in the exercise variables relate to changes in symptom-based, echocardiographic, and biomarker end points by correlation and multivariate regression analysis.

Baseline data are presented as numbers and percentages, means and SDs, and medians with IQRs. *P* values <.0031 were considered statistically significant based on 16 variables analyzed (0.05/16). Statistical analyses were performed by the Brigham and Women's Hospital Clinical Trials Outcomes Center (I.J.K. and B.L.C.) using STATA version 19 (StataCorp).

Results

Participant Population

Between June 20, 2023, and August 30, 2024, 175 participants (mean [SD] age, 57.7 [13.2] years; 73 [41.7%] female and 102 [58.3%] male) were randomized at 71 sites in North America, South America, Europe, Israel, and China, with the last patient/last visit in March 2025. Baseline characteristics were generally similar between the 2 treatment groups (**Table 1**). A total of 165 (94%) had core laboratory–validated exercise tests at baseline and week 24.

Patient Disposition

At least 1 dose of study medication was received by all randomized participants (eFigure 1 in Supplement 3). At week 24, there were 5 (5.7%), 15 (17.2%), 35 (40.2%), and 31 (35.6%) patients taking 5, 10, 15, and 20 mg of aficamten, respectively, with 1 (1.1%) participant's dose down-titrated to placebo according to the titration algorithm. In the metoprolol arm, 12 (14.3%), 10 (11.9%), 19 (22.6%), 34 (40.5%) patients were taking 50, 100, 150, and 200 mg of metoprolol, respectively, with the dose for 10 of them (11.9%) down-titrated to placebo (1 due to a transcription error). Core laboratory–validated exercise tests at baseline and week 24 were available for 165 randomized trial participants (94%). Ten participants (6%; 5 in aficamten and 5 in metoprolol) did not have a valid week 24 CPET available for analysis. Of these, 1 participant in the aficamten

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Table 1. Demographic and Clinical Characteristics of Participants at Baseline[a]

Characteristic	Aficamten (n = 88)	Metoprolol (n = 87)
Age, mean (SD), y	58.9 (13)	56.5 (13)
Sex, No. (%)		
Male	52 (59.1)	50 (57.5)
Female	36 (40.9)	37 (42.5)
Race, No. (%)[b]		
Asian	13 (14.8)	12 (13.8)
Black or African American	1 (1.1)	0
White	70 (79.5)	70 (80.5)
Other[c]	4 (4.5)	5 (5.7)
Region, No. (%)		
China	11 (12.5)	11 (12.6)
North America	45 (51.1)	39 (44.8)
Europe, Israel, or Brazil	32 (36.4)	37 (42.5)
HCM history, No. (%)		
Family history	16 (18.2)	18 (20.7)
Pathogenic sarcomere variant	27 (30.7)	22 (25.3)
Time since HCM diagnosis, mean (SD), y	5.7 (6.61)	4.9 (5.74)
Medical history, No. (%)		
Hypertension	54 (61.4)	33 (37.9)
Diabetes	6 (6.8)	10 (11.5)
Coronary artery disease	8 (9.1)	12 (13.8)
Myocardial infarction	0	1 (1.1)
Sustained ventricular tachycardia	2 (2.3)	1 (1.1)
Background SoC therapy, No. (%)		
β-Blocker	64 (72.7)	59 (67.8)
Calcium channel blocker	12 (13.6)	10 (11.5)
Not on background therapy	19 (21.6)	23 (26.4)
Implantable cardioverter-defibrillator	8 (9.1)	13 (14.9)
Vital signs, mean (SD)		
Systolic blood pressure, mm Hg	125 (14)	126 (14)
Diastolic blood pressure, mm Hg	77 (10)	77 (9)
Heart rate, bpm	78 (12)	71 (12)
BMI	28.2 (3)	28.3 (5)
Symptoms		
KCCQ-CSS, mean (SD)	65.5 (17)	66.0 (16)
NYHA functional class, No. (%)		
Class II	63 (72)	60 (69)
Class III	25 (28)	27 (31)
Biomarkers, median (IQR)		
NT-proBNP, pg/mL	510 (213-993)	439 (171-907)
hs-cTnI, ng/L	14.2 (7.3-34.0)	11.6 (6.3-25.1)
CPET, mean (SD)		
Peak RER	1.17 (0.08)	1.19 (0.10)
Peak VO_2, mL/kg/min	19.5 (4.8)	20.2 (5.3)
% Predicted pVO_2	60.0 (13.5)	61.2 (13.7)
Echo core laboratory, mean (SD)		
LVEF %	68.3 (3.8)	67.3 (3.9)
Resting LVOT-G, mm Hg	48.6 (30.1)	46.2 (27.4)
Valsalva LVOT-G, mm Hg	75.3 (34.0)	71.6 (31.2)
Maximum wall thickness, mm	20.9 (2.8)	20.8 (3.3)
LA volume index, mL/m^2	37.8 (10.9)	40.1 (12.1)

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); CPET, cardiopulmonary exercise testing; echo, echocardiogram; hs-cTnI, high-sensitivity cardiac troponin I; KCCQ-CSS, Kansas City Clinical Questionnaire clinical summary score; LA, left atrial; LVOT-G, left ventricular output tract gradient; NT-proBNP, N-terminal pro–B-type natriuretic peptide; NYHA, New York Heart Association; oHCM, obstructive hypertrophic cardiomyopathy; pVO_2, peak oxygen uptake; RER, respiratory exchange ratio; SoC, standard of care; VO_2, oxygen uptake.

[a] Percentages may not total 100 because of rounding.

[b] Race and ethnicity data were collected via self-report and included to classify the diversity of the sample.

[c] Other race and ethnicity groups included another race or ethnicity, not listed, or not known, consolidated because of small numbers.

arm died, 1 participant in the metoprolol arm experienced early study termination related to an adverse event prior to the week 24 CPET, and 8 (3.4%; 4 in aficamten and 4 in metoprolol) were determined by the core laboratory to have an invalid week 24 CPET (6 were technical failures, 1 had a transient condition that interfered with exercise testing, and in 1 case, the site did not follow the CPET manual of operations) (eFigure 1 in Supplement 3).

Exercise End Points

Submaximal Exercise Measures

Compared with metoprolol, aficamten improved submaximal exercise performance as indicated by the divergent changes in VO_2 at the ventilatory anaerobic threshold (76 mL; 95% CI, 41 to 111; $P < .001$) and VE/VCO_2 slope throughout exercise (−2.8 mL; 95% CI, −4.0 to −1.5; $P < .001$). Aficamten treatment also tended to improve ability to perform aerobic work based on the higher VO_2/work slope (0.8 mL/min/watt; 95% CI, 0.2 to 1.3; $P = .004$) and preanaerobic threshold ventilatory efficiency (VE/VCO_2 slope: −1.3; 95% CI, −2.3 to −0.3; $P = .01$), though these 2 measurements did not reach criteria for statistical significance after Bonferroni correction of our P value threshold for significance (**Table 2**).

Peak Exercise Measures

The peak respiratory exchange ratios were nearly identical between baseline and week 24 for both aficamten and metoprolol groups, reflecting adequate volitional performance. Compared with metoprolol, aficamten improved peak exercise measures, including pVO_2 (2.3 mL/kg/min; 95% CI, 1.5 to 3.1; $P < .001$), peak workload (8 watts; 95% CI, 3 to 13; $P = .003$), time required for VO_2 to recover by 12.5% postexercise (−11 seconds; 95% CI, −16 to −5; $P < .001$), circulatory power (819 mm Hg × mL/min/kg; 95% CI, 569 to 1070; $P < .001$), total exercise duration (0.6 min; 95% CI, 0.2 to 1.0; $P = .002$), and HR

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Table 2. Cardiopulmonary Exercise Testing Parameters by Treatment Assignment

Variable	Aficamten				Metoprolol				Between-group difference (95% CI)	*P* value
	No.	Baseline, mean (SD)	Week 24, mean (SD)	Absolute difference (SD)	No.	Baseline, mean (SD)	Week 24, mean (SD)	Absolute difference (SD)		
Submaximal exercise response variables										
Anaerobic threshold, mL	83	924 (250)	961 (259)	37 (122)	81	1004 (314)	960 (307)	−44 (106)	76 (41 to 111)	<.001
Aerobic efficiency (VO_2/work), mL/min/watt	83	9.2 (2.2)	9.5 (2.3)	0.3 (2.0)	80	9.6 (2.3)	9.0 (2.2)	−0.6 (1.8)	0.8 (0.2 to 1.3)	.004
Ventilatory efficiency (preanaerobic threshold), VE/VCO_2 slope	82	29.5 (4.4)	27.6 (3.8)	−1.9 (4.2)	81	29.2 (4.8)	28.7 (4.4)	−0.5 (3.7)	−1.3 (−2.3 to −0.3)	.01
Ventilatory efficiency, VE/VCO_2 slope	83	33.8 (6.4)	31.1 (4.8)	−2.8 (5.4)	82	33.4 (5.8)	33.6 (6.5)	0.2 (3.5)	−2.8 (−4.0 to −1.5)	<.001
Peak exercise response variables										
Peak VO_2 per kg, mL/kg/min	83	19.6 (4.6)	20.7 (5.0)	1.1 (2.8)	82	20.3 (5.4)	19.0 (5.7)	−1.2 (2.2)	2.3 (1.5 to 3.1)	<.001
Peak workload, watts	82	119 (41)	126 (43)	7 (16)	82	119 (45)	118 (45)	−1 (17)	8 (3 to 13)	.003
Peak heart rate, bpm	82	149 (17)	154 (17)	5 (11)	82	151 (20)	127 (21)	−23 (16)	28 (24 to 32)	<.001
Exercise duration, min	79	11.7 (2.9)	12.2 (3.1)	0.5 (1.2)	78	11.7 (3.1)	11.7 (3.2)	−0.1 (1.3)	0.6 (0.2 to 1.0)	.002
Heart rate reserve, bpm	82	66 (20)	71 (20)	5 (12)	80	69 (20)	62 (19)	−7 (14)	12 (8 to 16)	<.001
Peak RER	83	1.17 (0.08)	1.18 (0.10)	0.01 (0.08)	82	1.19 (0.11)	1.19 (0.11)	0.00 (0.10)	0.001 (−0.026 to 0.027)	.96
Postexercise recovery variables										
VO_2 recovery delay, >0%, s	82	16 (22)	12 (16)	−4 (21)	78	15 (19)	19 (20)	4 (22)	−7 (−12 to −2)	.009
VO_2 recovery 12.5%, s	77	36 (22)	31 (18)	−6 (19)	77	33 (19)	39 (21)	7 (19)	−11 (−16 to −5)	<.001
VO_2 recovery 25%, s	76	58 (21)	53 (18)	−5 (18)	76	50 (17)	57 (19)	7 (16)	−8 (−13 to −3)	.002
VO_2 recovery 50%, s	75	96 (37)	86 (23)	−10 (31)	72	82 (23)	93 (41)	11 (33)	−14 (−24 to −5)	.004
Composite exercise response variables										
Hemodynamic (SBP) + O_2 uptake: circulatory power, mm Hg × mL/min/kg	83	3413 (1116)	3782 (1273)	369 (993)	81	3439 (1131)	2993 (1013)	−446 (738)	819 (569 to 1070)	<.001
Hemodynamic + ventilatory efficiency, ventilatory power, mm Hg	82	5.3 (1.6)	6.0 (1.4)	0.7 (1.3)	81	5.2 (1.4)	4.9 (1.3)	−0.4 (1.1)	1.1 (0.8 to 1.4)	<.001
O_2 uptake + ventilatory efficiency, standardized 2-component *z* score	83	−0.05 (0.80)	0.18 (0.67)	0.23 (0.57)	82	0.06 (0.75)	−0.18 (0.81)	−0.24 (0.38)	0.45 (0.31 to 0.59)	<.001

Abbreviations: MET, metabolic equivalent; O_2, oxygen; RER, respiratory exchange ratio; VCO_2, carbon dioxide output; VE, minute ventilation; VO_2, oxygen uptake.

[a] The absolute difference corresponds to the change from baseline to week 24.

[b] The adjusted difference corresponds to the least-squares mean treatment difference.

[c] Integrated exercise performance was defined as the 2-component *z* score of peak VO_2 and ventilatory efficiency (VE/VCO_2 slope). The *z* score was derived by reversing the directionality of VE/VCO_2 slope values such that increases in both *z* score components indicate benefit; equal weights were used for each component.

reserve (12 bpm; 95% CI, 8 to 16; $P < .001$) (Table 2). There was consistent improvement in all peak-performance metrics with aficamten treatment, but deterioration with metoprolol treatment.

Recovery Exercise Measures

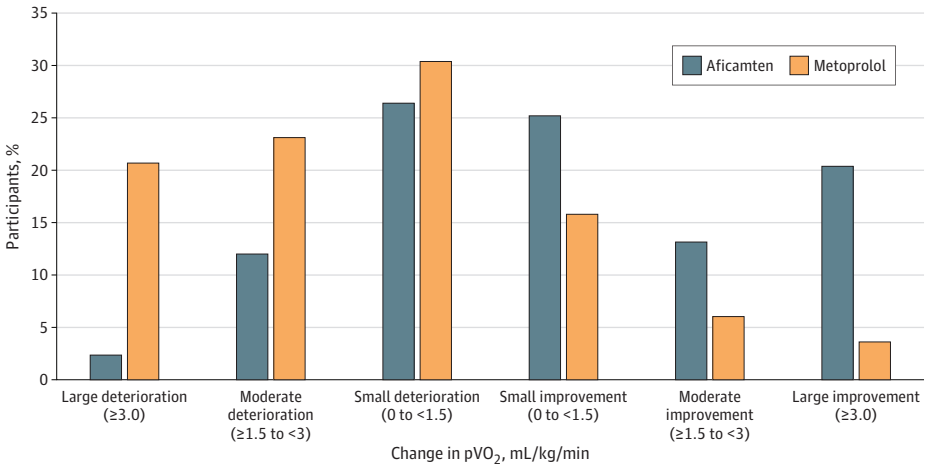
After cessation of exercise, VO_2 recovery time improved with aficamten compared with metoprolol, across all 4 time points assessed, encompassing the initial descent in VO_2 from peak as well as 12.5%, 25%, and 50% of VO_2 recovery to resting values. At each interval, the speed of VO_2 recovery increased with aficamten and slowed with metoprolol, with between-group difference exceeding 30% of baseline values for 12.5% recovery and remaining statistically significant after adjustment for multiple hypothesis testing at the 12.5% and 25% time points (Table 2).

Integrative Measures of Exercise Performance

We constructed a single composite exercise performance score that simultaneously captures both submaximal and peak exercise performance intended to reflect exercise per-

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Figure 1. Bar Graph Showing Responder Analyses of Categorical Change in Peak Oxygen Uptake (pVO_2) With Aficamten vs Metoprolol at 24 Weeks



Data represented as proportion of participants experiencing small (0 to <1.5), moderate (≥1.5 to <3), or large (≥3.0) improvements or deteriorations in pVO_2 (mL/kg/min), per treatment group. NNH indicates number needed to harm with metoprolol as the reference group; NNT indicates number needed to treat with aficamten as the reference group; OR, odds ratio.

formance across the spectrum of activity. This composite exercise performance score demonstrated improvement with the aficamten treatment (*z* score mean [SD], 0.23 [0.57]) compared with a deterioration in the metoprolol group (*z* score mean [SD], −0.24 [0.38]), yielding a significant between-group increase (*z* score mean, 0.45; 95% CI, 0.31 to 0.59; $P < .001$) (Table 2). Aficamten also improved composite variables combining gas exchange and systolic blood pressure measures, including circulatory power and ventilatory power (Table 2). The mean (SD) peak systolic BP component variable of circulatory power increased with aficamten (9 [32] mm Hg) relative to metoprolol (−12 [27] mm Hg), with a between-group difference of 24 (95% CI, 16 to 31 mm Hg; $P < .001$).

Responder Analysis

Large improvements (≥3.0 mL/kg/min) in pVO_2 were significantly more common with aficamten compared with metoprolol (17 [20.5%] vs 3 [3.7%], respectively; odds ratio [OR], 6.8; 95% CI, 2.0 to 22.5; $P < .001$). In contrast, large reductions (≥3.0 mL/kg/min) were significantly more common with metoprolol than aficamten (17 [20.7%] vs 2 [2.4%], respectively; OR, 10.6; 95% CI, 2.6 to ∞; $P < .001$) (**Figure 1**).

The ORs for achieving any improvement (OR, 4.19; 95% CI, 2.17 to 8.08) or moderate to large improvements (≥1.5 mL/kg/min) (OR, 4.71; 95% CI, 2.02 to 10.92), also favored aficamten treatment and correspond with a number needed to treat of 3.0 and 4.2 participants, respectively (eTable 3 in Supplement 3). Conversely, the OR for achieving any worsening, moderate to large worsening, or large worsening in pVO_2 were higher in the metoprolol arm with number needed to harm below 6 for all 3 strata (number needed to harm: 3.0, 3.4, and 5.5 for any, moderate to large, and large worsening respectively). Combining large improvements and avoidance of large worsening in exercise capacity, the number needed to treat for aficamten vs metoprolol was 2.8.

Clinical Correlations

We identified significant associations between improvements in CPET variables and improvements in Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score (KCCQ-CSS) and NYHA functional class, with changes in the composite *z* score most closely correlated with both (**Figure 2**; eFigure 2 in Supplement 3). Among individual gas exchange variables, VO_2 recovery (T12.5%) was the only variable for which changes were significantly associated with changes in all functional status/quality of life, echocardiographic, and NT-proBNP changes (Figure 2; eFigure 2 in Supplement 3). Sequential regression multivariate analyses revealed that changes in NT-proBNP accounted for the greatest amount of variance in change in pVO_2 (eTable 2 in Supplement 3).

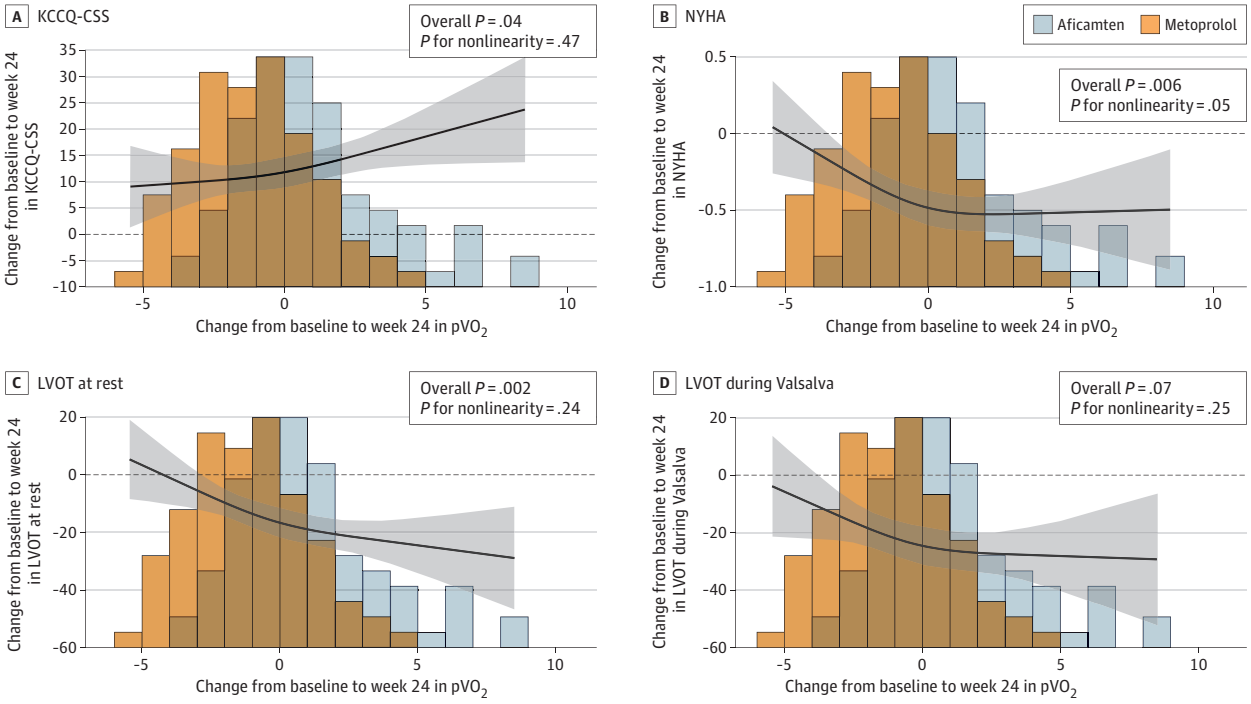
To further investigate changes in pVO_2 in response to aficamten or metoprolol, we evaluated changes in its 2 component variables, peak HR and peak oxygen (O_2) pulse (VO_2/HR). Both component variables changed concordantly with pVO_2 by cubic spline analysis (**Figure 3**) but notably O_2 pulse remained relatively unchanged across the spectrum of changes in pVO_2 at increments below 3 mL/kg/min, with greater increments in O_2 pulse among participants predominantly treated with aficamten who achieved increments in pVO_2 greater than 3 mL/kg/min (Figures 1 and 3). In contrast, participants with a fall in pVO_2 greater than 3 mL/kg/min were predominantly composed of patients who received metoprolol (Figure 1), with corresponding reductions from baseline in peak HR greater than 20 bpm, as illustrated by cubic spline analysis (Figure 3).

Discussion

This prespecified analysis of a randomized clinical trial of exercise measures across multiple stages of activity demonstrated divergent physiologic responses to exercise with aficamten monotherapy compared with metoprolol in participants with symptomatic oHCM. Aficamten conferred ben-

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Figure 2. Cubic Spline Graphs Showing the Correlation Between Change in Peak Oxygen Uptake (pVO_2) and Clinical Metrics



Change in KCCQ-CSS (A), NYHA functional class (B), resting LVOT-G (C), and Valsalva LVOT-G (D) in relation to change in pVO_2 from baseline to 24 weeks. Linear regression with cubic terms. Analyses adjusted for baseline values of both parameters. Solid lines and shading show the association correlate with 95% CIs. Histograms show the distribution of change in pVO_2. KCCQ-CSS indicates Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score; LVOT-G, left ventricular outflow tract gradient; NYHA, New York Heart Association; pVO_2, peak oxygen uptake.

efits throughout submaximum and maximum exercise, augmenting exercise duration, oxygen uptake, ventilatory efficiency, workload achieved, and hemodynamic measures, including circulatory and ventilatory power, while also speeding up VO_2 recovery (T12.5% and T25%). These findings provide compelling justification for aficamten treatment as an appropriate alternative to current first-line treatment with metoprolol, to treat exercise intolerance in oHCM.

Targeting Exercise Intolerance in oHCM

In oHCM cardiovascular event rates are infrequent but exercise intolerance is extremely common, with a meta-analysis[5] of 60 studies that measured pVO_2 in HCM revealing an average pVO_2 in oHCM that only modestly exceeds what is required to support independent living. As a result, exercise capacity is a critically important therapeutic target in oHCM. In our categorical threshold analyses, aficamten participants were more than 6 times as likely to achieve a large pVO_2 increase (≥3.0 mL/kg/min) and more than 10 times less likely to experience large worsening than those receiving metoprolol (Figure 1). These findings are particularly noteworthy given that exercise capacity is closely related to outcomes in HCM, with a 1 mL/kg/min higher pVO_2 associated with a 14% lower rate of mortality and 13% lower composite adverse CV outcomes in a recent meta-analysis of CPET studies in HCM.[5]

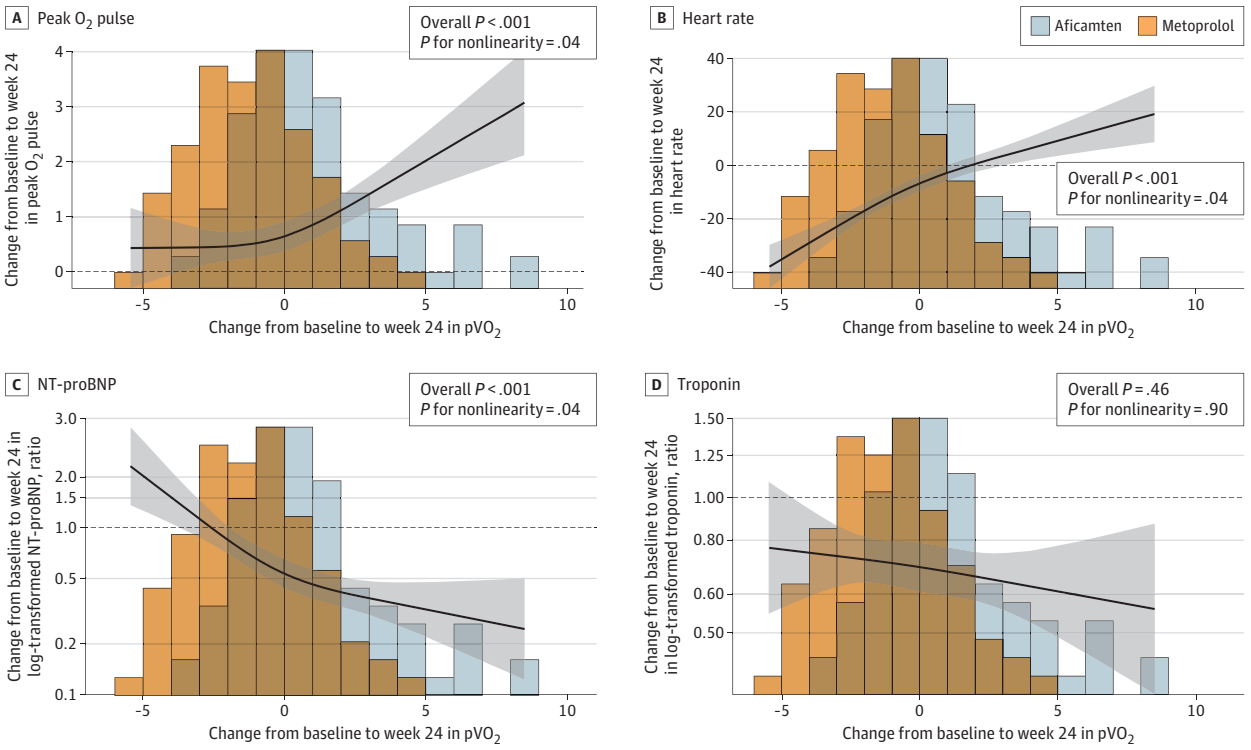
Impact of Heart Rate Augmentation During Exercise in oHCM

Dybro et al[18] examined metoprolol vs placebo in a randomized, double-blind crossover study with invasive hemodynamic monitoring during exercise that permitted dissection in VO_2 component variable responses to exercise. Their study highlighted the asymmetric relative augmentation patterns in stroke volume (9.8%) vs HR (80.5%) from rest to peak exercise at baseline. Therefore, even though peak exercise stroke volume increased modestly with metoprolol, the >20% reduction in the more dynamic HR variable ultimately resulted in a net reduction in peak exercise cardiac output of approximately 2 L/min.[18] Furthermore, metoprolol did not improve exercise pulmonary capillary wedge pressure, suggesting a net deleterious hemodynamic impact of β-blockade on peak exercise cardiac performance in oHCM. Our noninvasive CPET study did not permit quantification of relative stroke volume and HR excursions with exercise, but the within-group analysis suggested metoprolol exposure in MAPLE-HCM lowered peak HR (−23 [16] bpm), whereas mean (SD) peak HR was 5 (11) bpm higher with aficamten treatment in MAPLE-HCM.

Increases in mean (SD) peak HR in the MAPLE-HCM aficamten group (5 [11] bpm) are similar to the observed increments in peak HR in the aficamten treatment group of

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Figure 3. Cubic Spline Graphs Showing the Correlation Between Changes in Peak Oxygen Uptake (pVO_2) and Component Variables of pVO_2 as Well as Biomarkers



Change in pVO_2 component variables, peak O_2 pulse (A), and peak heart rate (B) in relation to change in pVO_2 from baseline to 24 weeks as well as change in pVO_2 relative to change in N-terminal pro–B-type natriuretic peptide (NT-proBNP) (C) and high-sensitivity cardiac troponin I (hs-cTnI) (D). Linear regression with cubic terms. Analyses adjusted for baseline values of both parameters. Solid lines and shading show the association correlate with 95% CIs. Histograms show the distribution of change in pVO_2.

SEQUOIA-HCM (8 [14] bpm).[15] Further, we observed correlations between changes in peak HR and changes in NT-proBNP as well as LVOT-G at rest and with Valsalva (eFigure 2 in Supplement 3). These findings likely reflect a hemodynamic environment, with lower LVOT and wall stress, in which HR was permitted to reach greater heights during maximum volitional effort exercise, rather than any direct impact of cardiac myosin inhibitor on sinus node activity. Indeed, as with other studies with aficamten in oHCM, there was no change in resting HR.[7] These findings highlight the clear difference between metoprolol and aficamten in preserving HR reserve during exercise, and emphasize the often-overlooked importance of chronotropic response when evaluating causes of exercise intolerance and treatment choices in oHCM

Our findings mirror recent trial results in heart failure with preserved ejection fraction. In heart failure with preserved ejection fraction, pharmacotherapies that lower HR during exercise have been historically postulated to permit hearts with diastolic dysfunction time to fill, yet Palau et al[19,20] found that β-blocker withdrawal increased pVO_2 by 2.1 mL/kg/min.

Impact of Aficamten vs Metoprolol on CPET Measures Beyond pVO_2

A frequent criticism of pVO_2 as an end point in clinical trials is that on a day-to-day basis individuals spend very little time at their peak VO_2, thereby making this assessment potentially less reflective of an impact on daily living. Pharmacotherapies, including β-blockers have been shown to exert distinct effects on different domains of exercise,[21,22] such as being able to tolerate longer duration of low-level exercise if specific mechanisms of exercise intolerance (eg, myocardial ischemia) are improved. Although early observational studies with β-blockade (propranolol) in oHCM did suggest longer achievable exercise duration due to amelioration of anginal symptoms[6] our data demonstrates that treatment with aficamten remains superior to metoprolol using measures of submaximal exercise. Improved VO_2 at anaerobic threshold, which occurs at exercise intensity levels approximately half of those achieved at peak exercise, indicates that submaximal exercise capacity is also improved by aficamten, as is the overall exercise time. In addition, aficamten improved measures spanning exercise that are relatively independent of HR, including the ability to perform work aerobically, to breathe efficiently throughout exercise, and to recover from exercise more rapidly, in comparison with metoprolol.

Use of CPET to Ascertain Broad Measures of Exercise Capacity

The MAPLE-HCM trial reflects successful scaling of CPET in a clinical trial that spanned 71 sites internationally and builds

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on the successful implementation of the SEQUOIA-HCM[9] and EXPLORER-HCM[23] trials. MAPLE-HCM has the broadest array of CPET variables measured in a multicenter clinical trial, providing insight into divergent effects of aficamten and metoprolol across multiple stages of exercise and recovery. High reproducibility and consistent achievement of maximum volitional effort (eg, peak respiratory exchange ratio values uniformly averaged more than 1.15 at baseline in both groups at baseline and week 24; Table 2) contribute to the power to detect differences in CPET measures. The established role of CPET in prognostication of HCM[24,25] and other cardiovascular diseases[26] lends further support for the use of CPET end points in clinical trials, including consideration of new end points such as the dual component pVO_2 and the VE/VCO_2 slope *z* score, which is a focal point of the phase 3 ACACIA-HCM trial (NCT06081894).

Limitations

This study has limitations. There were 10 participants (5.7%) (5 in each group) whose data were not included in this analysis due to technical issues, deviations from the CPET protocol, or adverse events, although the study was powered under the assumption that 10% or less of CPET data would be missing. These data do not inform long-term outcomes of either metoprolol or aficamten treatment of oHCM, as the study duration was 24 weeks. Further investigation is also needed to determine the effect of aficamten on exercise responses in asymptomatic patients with oHCM.

Conclusions

The findings of this randomized clinical trial included improved exercise response patterns in participants with oHCM treated with aficamten compared with metoprolol across all phases of exercise, including HR-independent measures. The divergent effects of monotherapy with aficamten or metoprolol on exercise performance lends support to consideration of aficamten as first-line monotherapy to augment functional capacity in patients with symptomatic oHCM.

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Group Information: The MAPLE-HCM Investigators are listed in Supplement 4.

Author Affiliations: Mass General Brigham Heart and Vascular Institute, Boston, Massachusetts (Lewis, McGinnis, Newlands); Hospital Universitario Puerta de Hierro de Majadahonda, Instituto de Investigación Sanitaria Puerta de Hierro (IDIPHISA), Centro de Investigación Biomédica en Red en Enfermedades Cardiovasculares (CIBERCV), Madrid, Spain (Garcia-Pavia); Centro Nacional de Investigaciones Cardiovasculares, Madrid, Spain (Garcia-Pavia); Oregon Health Science University, Portland (Masri); Heart and Vascular Center, Semmelweis University, Budapest, Hungary (Merkely); University of Missouri Kansas City Healthcare Institute for Innovations in Quality and Saint Luke's Mid America Heart Institute, Kansas City (Nassif); General Hospital of H.U. Virgen Del Rocio, Seville, Spain (Peña-Peña); Complexo Hospitalario Universitario A Coruña, Instituto de Investigación Biomédica de A Coruña (INIBIC), Centro de Investigación Biomédica en Red Enfermedades Cardiovasculares-Cardiovascular-Instituto de Salud Carlos III (CIBERCV-ISCIII), A Coruña, Spain (Barriales-Villa); Emory University Medical Center, Atlanta, Georgia (Bilen); Wellstar Center for Cardiovascular Care, Hiram, Georgia (Burroughs); Cardiovascular Division, Brigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts (Claggett, Kulac, Lakdawala, Solomon); Instituto Cardiovascular de Buenos Aires, Buenos Aires, Argentina (Costabel); Dante Pazzanese Institute of Cardiology, São Paulo, Brazil (de Barros Correia); Aarhus Hospital University, Aarhus, Denmark (Dybro, Poulsen); Patient Author, Colorado Springs, Colorado (Mann); Lahey Hospital and Medical Center, Burlington, Massachusetts (Maron); Texas Heart Institute at Baylor College of Medicine, Houston (Nair); University Hospital Center of Bordeaux, CIC-P 1401, University of Bordeaux, Bordeaux, France (Reant); University Hospital Jena, Friedrich-Schiller-University, Jena, Germany (Schulze); Duke University Medical Center, Durham, North Carolina (Wang); Cytokinetics, South San Francisco, California (Sohn, Berhane, Heitner, Jacoby, Kupfer, Malik, Wohltman); Massachusetts General Hospital, Harvard Medical School, Boston (Fifer); Department of Inherited Cardiovascular Diseases, St Bartholomew's Hospital, Institute of Cardiovascular Science, University College London, London, United Kingdom (Elliott).

Author Contributions: Drs Lewis and Claggett had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Lewis, Masri, Bilen, Maron, Nair, Solomon, Heitner, Jacoby, Kupfer, Malik, Wohltman, Fifer, Elliott.
Acquisition, analysis, or interpretation of data: Lewis, Garcia-Pavia, Masri, Merkely, Nassif, Peña-Peña, Barriales-Villa, Burroughs, Claggett, Costabel, de Barros Correia, Dybro, Kulac, Lakdawala, Mann, Lyn McGinnis, Newlands, Poulsen, Reant, Schulze, Wang, Sohn, Berhane, Heitner, Jacoby, Kupfer, Malik, Wohltman.
Drafting of the manuscript: Lewis, Kulac, Lyn McGinnis, Poulsen, Reant, Schulze, Heitner, Fifer, Elliott.
Critical review of the manuscript for important intellectual content: Lewis, Garcia-Pavia, Masri, Merkely, Nassif, Peña-Peña, Barriales-Villa, Bilen, Burroughs, Claggett, Costabel, de Barros Correia, Dybro, Lakdawala, Mann, Maron, Nair, Newlands, Poulsen, Reant, Schulze, Solomon, Wang, Sohn, Berhane, Heitner, Jacoby, Kupfer, Malik, Wohltman, Elliott.
Statistical analysis: Lewis, Bilen, Claggett, Kulac, Solomon, Berhane, Heitner, Wohltman.
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Supervision: Lewis, Masri, Peña-Peña, Barriales-Villa, Bilen, Claggett, de Barros Correia, Dybro, Mann, Maron, Poulsen, Schulze, Solomon, Jacoby, Kupfer, Malik, Fifer, Elliott.

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